

## Is this proof of the linear convergence of Newton's method correct?

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**Newton's root finding method** has at least quadratic convergence rate when used to find a **simple root**  $x^*$  of a function  $f(x)$  (by simple root I mean that  $f(x^*) = 0$ , but every derivative of  $f$  evaluated at  $x^*$  is non-zero).

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However, if  $x^*$  has multiplicity  $m > 1$ , then the convergence rate becomes **linear**.



Let  $g(x) = x - \frac{f(x)}{f'(x)}$ . I know that, if  $g'(x) \neq 0$  (which implies that  $x^*$  is not a simple root), then the following result holds:



$$\lim_{k \rightarrow +\infty} \frac{|x_{k+1} - x^*|}{|x_k - x^*|} = |g'(x^*)| < 1$$

this means that the convergence rate is linear (even though I have no clue about where this result comes from).

I tried to give a proof to show that the convergence rate is, indeed, linear when  $x^*$  is a root of multiplicity greater than 1.

### My proof

Using Taylor's expansion, i rewrite  $f(x)$  as

$$f(x) = f(x^*) + f'(x^*)(x - x^*) + \frac{f''(x^*)}{2!}(x - x^*)^2 + \dots + \frac{f^{(m)}(x^*)}{m!}(x - x^*)^m + \dots$$

Since  $x^*$  has multiplicity  $m$ , the first  $m$  terms are equal to zero

$$f(x) = \cancel{f(x^*)} + \cancel{f'(x^*)(x - x^*)} + \cancel{\frac{f''(x^*)}{2!}(x - x^*)^2} + \dots + \frac{f^{(m)}(x^*)}{m!}(x - x^*)^m + \dots$$

$$f(x) = \frac{f^{(m)}(x^*)}{m!} \cdot (x - x^*)^m + O((x - x^*)^{m+1})$$

It's first derivative is

$$f'(x) = \frac{f^{(m)}(x^*)}{m!} \cdot m(x - x^*)^{m-1} + O((x - x^*)^m)$$

Now, substituting these expressions into Newton's iteration formula:

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)} = x_k - \frac{\frac{f^{(m)}(x^*)}{m!} \cdot (x_k - x^*)^m + O((x_k - x^*)^{m+1})}{\frac{f^{(m)}(x^*)}{m!} \cdot m(x_k - x^*)^{m-1} + O((x_k - x^*)^m)}$$

As  $x_k \rightarrow x^*$ , the term  $O((x_k - x^*)^m)$  becomes negligible

$$x_{k+1} \approx x_k - \frac{\cancel{\frac{f^{(m)}(x^*)}{m!}} \cdot (x_k - x^*)^m + \cancel{O((x_k - x^*)^m)}}{\cancel{\frac{f^{(m)}(x^*)}{m!}} \cdot m(x_k - x^*)^{m-1} + \cancel{O((x_k - x^*)^m)}}$$

$$x_{k+1} \approx x_k - \frac{(x_k - x^*)^m}{m(x_k - x^*)^{m-1}}$$

$$x_{k+1} \approx x_k - \frac{(x_k - x^*)}{m}$$

$$x_{k+1} - x^* \approx x_k - x^* - \frac{(x_k - x^*)}{m}$$

$$e_{k+1} \approx e_k - \frac{e_k}{m}$$

Where  $e_k$  is the error  $(x_k - x^*)$  at step  $k$ , and  $e_{k+1}$  is the error at step  $k+1$

$$e_{k+1} \approx e_k \left(1 - \frac{1}{m}\right)$$

$$|e_{k+1}| \approx |e_k| \left(\frac{m-1}{m}\right) \quad \square$$

The result seems correct. Also, it makes sense that, as  $m$  grows, the convergence rate becomes slower but remains linear. I'm not so sure about the whole process though. Is the result correct? Does the proof contain errors? Thank you.

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asked 22 mins ago

 BigBuckBunny  
1 1

 New contributor

 It is correct, the big O should be  $O(|x - x^*|^{m+1})$  in the numerator. – Yirmin 12 mins ago